

The polynomial is monic degree 7, so write

$$P(x) = x^7 + a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0,$$

with integer coefficients a_i . Conditions:

- a_i are odd, $|a_i| < 2025^2$ for $i = 0, \dots, 6$; leading coefficient 1 also satisfies $|1| = 1$ odd and $< 2025^2$.
- For each coefficient (including the leading one) its absolute value is a root of P . In particular 1 (absolute value of leading coefficient) is a root, so $P(1) = 0$:

$$1 + \sum_{i=0}^6 a_i = 0 \implies \sum_{i=0}^6 a_i = -1.$$

Define $t_i = |a_i| > 0$ (odd). Then the condition gives $P(t_i) = 0$ for each $i = 0, \dots, 6$ as well. By the rational root theorem any integer root divides the constant term, hence $t_i \mid a_0$ for all i . Therefore $t_0 = |a_0|$ and each t_i is a positive odd divisor of t_0 .

We split into two cases.

0.0.1 Case 1: $t_0 = 1$

Then $a_0 = \pm 1$ and every $t_i \mid 1$ forces $t_i = 1$. Hence all coefficients (except the leading 1) are ± 1 . Equation (1) becomes $1 + \sum_{i=0}^6 \varepsilon_i = 0$, where $\varepsilon_i = \pm 1$ is the sign of a_i . Thus $\sum_{i=0}^6 \varepsilon_i = -1$, i.e. among the seven coefficients $a_0, \dots, a_6$ exactly three are $+1$ and four are -1 . The number of such sign choices is $\binom{7}{3} = 35$. Each such polynomial satisfies $P(1) = 0$ (so 1 is a root) and every coefficient has absolute value 1 which is a root; the other conditions (odd, $< 2025^2$) are clearly satisfied. Hence there are 35 good polynomials with $t_0 = 1$.

0.0.2 Case 2: $t_0 > 1$

Now t_0 is odd and > 1 . Write $a_i = \varepsilon_i t_i$ with $\varepsilon_i = \pm 1$ (so $t_i = |a_i|$). The conditions give

$$\sum_{i=0}^6 \varepsilon_i t_i = -1,$$

$$\sum_{i=0}^6 \varepsilon_i t_i t_0^i = -t_0^7.$$

Because $t_i \mid t_0$, each t_i is either 1 or a proper divisor of t_0 or t_0 itself. Manipulate (3) step by step using increasing powers of t_0 to show that the only possibility is: $t_i = t_0$ for even i and $t_i = 1$ for odd i , together with precise sign relations.

Step 1: Divide (3) by t_0 :

$$\varepsilon_0 + \varepsilon_1 t_1 + \sum_{i=2}^6 \varepsilon_i t_i t_0^{i-1} = -t_0^6.$$

Modulo t_0 the sum for $i \geq 2$ vanishes; thus $\varepsilon_0 + \varepsilon_1 t_1 \equiv 0 \pmod{t_0}$. Since $t_1 \mid t_0$ and $t_0 > 1$, the only possibility is $t_1 = 1$ and $\varepsilon_1 = -\varepsilon_0$. (Otherwise $|t_1| > 1$ gives a contradiction because $|\varepsilon_0 + \varepsilon_1 t_1|$ would be smaller than t_0 and not a multiple of t_0 .) Hence $t_1 = 1$, $\varepsilon_1 = -\varepsilon_0$.

Step 2: Insert this into (3). The terms for $i = 0, 1$ become $\varepsilon_0 t_0 + (-\varepsilon_0) t_0 = 0$. So (3) reduces to

$$\sum_{i=2}^6 \varepsilon_i t_i t_0^i = -t_0^7.$$

Divide by t_0 :

$$\varepsilon_2 t_2 + \varepsilon_3 t_3 t_0 + \varepsilon_4 t_4 t_0^2 + \varepsilon_5 t_5 t_0^3 + \varepsilon_6 t_6 t_0^4 = -t_0^5.$$

Reduce (4) modulo t_0 : the terms for $i \geq 3$ contain a factor t_0 , so $\varepsilon_2 t_2 \equiv 0 \pmod{t_0}$. As before, because $t_2 \mid t_0$ and $t_0 > 1$, this forces $t_2 = t_0$ (and ε_2 free). Thus $t_2 = t_0$.

Step 3: Substitute $t_2 = t_0$ into (4) and divide by t_0 :

$$\varepsilon_2 + \varepsilon_3 t_3 + \varepsilon_4 t_4 t_0 + \varepsilon_5 t_5 t_0^2 + \varepsilon_6 t_6 t_0^3 = -t_0^4.$$

Modulo t_0 we get $\varepsilon_2 + \varepsilon_3 t_3 \equiv 0 \pmod{t_0}$. Arguing as before, this forces $t_3 = 1$ and $\varepsilon_3 = -\varepsilon_2$. (Any proper divisor would give a non-zero residue of absolute value less than t_0 ; $t_3 = t_0$ would lead to $\varepsilon_2 \equiv 0$ impossible.) Hence $t_3 = 1$, $\varepsilon_3 = -\varepsilon_2$.

Step 4: With these values the left side of (5) simplifies: $\varepsilon_2 + \varepsilon_3 t_3 = 0$. Dividing the rest by t_0 gives

$$\varepsilon_4 t_4 + \varepsilon_5 t_5 t_0 + \varepsilon_6 t_6 t_0^2 = -t_0^3.$$

Reduce (6) modulo t_0 : $\varepsilon_4 t_4 \equiv 0 \pmod{t_0} \Rightarrow t_4 = t_0$ (and ε_4 free). Thus $t_4 = t_0$.

Step 5: Put $t_4 = t_0$ into (6) and divide by t_0 :

$$\varepsilon_4 + \varepsilon_5 t_5 + \varepsilon_6 t_6 t_0 = -t_0^2.$$

Modulo t_0 : $\varepsilon_4 + \varepsilon_5 t_5 \equiv 0 \pmod{t_0} \Rightarrow t_5 = 1$, $\varepsilon_5 = -\varepsilon_4$. So $t_5 = 1$, $\varepsilon_5 = -\varepsilon_4$.

Step 6: Then (7) reduces to $\varepsilon_4 + \varepsilon_5 t_5 = 0$, leaving

$$\varepsilon_6 t_6 t_0 = -t_0^2 \implies \varepsilon_6 t_6 = -t_0.$$

Since $t_6 \mid t_0$ and $t_0 > 1$, we must have $t_6 = t_0$ and $\varepsilon_6 = -1$. Thus $t_6 = t_0$, $\varepsilon_6 = -1$.

Collecting the results:

$$\begin{aligned} t_0 &= t_0, \quad t_1 = 1, \quad t_2 = t_0, \quad t_3 = 1, \quad t_4 = t_0, \quad t_5 = 1, \quad t_6 = t_0; \\ \varepsilon_1 &= -\varepsilon_0, \quad \varepsilon_3 = -\varepsilon_2, \quad \varepsilon_5 = -\varepsilon_4, \quad \varepsilon_6 = -1. \end{aligned}$$

So the absolute values are fixed: even-indexed coefficients have absolute value t_0 , odd-indexed have absolute value 1. The signs $\varepsilon_0, \varepsilon_2, \varepsilon_4$ are still free; the others are determined by them.

Now impose the sum condition (2):

$$\sum_{i=0}^6 \varepsilon_i t_i = \varepsilon_0 t_0 + (-\varepsilon_0) + \varepsilon_2 t_0 + (-\varepsilon_2) + \varepsilon_4 t_0 + (-\varepsilon_4) + (-1) t_0 = (T - 1) t_0 - T,$$

where $T = \varepsilon_0 + \varepsilon_2 + \varepsilon_4$. Setting this equal to -1 yields

$$(T - 1) t_0 - T = -1 \implies (T - 1)(t_0 - 1) = 0.$$

Since $t_0 > 1$, we must have $T = 1$. Hence $\varepsilon_0 + \varepsilon_2 + \varepsilon_4 = 1$, i.e. exactly two of them are $+1$ and one is -1 . That gives 3 possibilities for the triple $(\varepsilon_0, \varepsilon_2, \varepsilon_4)$.

Therefore, for every odd t_0 with $1 < t_0 < 2025^2$, there are exactly 3 good polynomials given by the above pattern of signs. The condition $|a_i| < 2025^2$ requires $t_0 < 2025^2$; the oddness

condition is already satisfied because t_0 is odd. The number of such t_0 is the number of odd integers from 3 to $2025^2 - 2$ inclusive, which is

$$\frac{2025^2 - 3}{2} = 2\,050\,311.$$

Hence the number of good polynomials from this case is $3 \times 2\,050\,311 = 6\,150\,933$.

Adding the 35 from the $t_0 = 1$ case gives the total

$$35 + 6\,150\,933 = 6\,150\,968.$$

6150968
